

Question:

Carlos is planting a lemon tree. The tree will cost \$90 to plant. Each year it will grow 7 lemons, which he can sell for \$1.5 each. It costs \$3 a year to water and feed the tree. How many years will it take before he starts earning money on the lemon tree?

Reasoning + Answer:

<tool\_call> Reasoning process here :

Step 1 : Understand the setup .

- Initial cost to plant the tree : \$ 9 0 ( one-time cost ).
- Each year , the tree produces 7 lemons .
- Each lemon sells for \$ 1 . 5 0 .
- Annual cost to water and feed : \$ 3 .
- We need to find the number of years until Carlos starts earning money , i . e . , when the income from lemons exceeds the annual expenses ( including the initial cost being amortized over time ).

Step 2 : Determine annual profit .

Each year :

- Revenue from lemons = 7 lemons Ã \$ 1 . 5 0 = \$ 1 0 . 5 0 .
- Cost to water and feed = \$ 3 .
- So , annual profit = \$ 1 0 . 5 0 - \$ 3 = \$ 7 . 5 0 .

But this is only after the initial \$ 9 0 cost . We need to determine when the cumulative profit from the tree ( after covering the initial cost ) becomes positive , i . e . , when the total income exceeds the total cost .

Letâs calculate the total cost after n years :

- Initial cost : \$ 9 0
- Annual cost ( water & feed ): \$ 3 per year â after n years : 3 n
- Total cost after n years = 9 0 + 3 n

Total income from lemons after n years :

- Each year : 7 Ã 1 . 5 = \$ 1 0 . 5 0 â total income = 1 0 . 5 n

We need to find the smallest integer n such that :

Total income > Total cost  
 $10.5n > 90 + 3n$

Step 3 : Solve the inequality

$10.5n > 90 + 3n$   
 $10.5n - 3n > 90$   
 $7.5n > 90$   
 $n > 90 / 7.5$   
 $n > 12$

So ,  $n > 12$  â the smallest integer n is 13 .

Now verify :

After 12 years :

- Income = 10 . 5 Ã 12 = \$ 126
- Cost = 90 + 3 Ã 12 = 90 + 36 = \$ 126 â break even , not earning money yet .

After 13 years :

- Income = 10 . 5 Ã 13 = 136 . 5
- Cost = 90 + 3 Ã 13 = 90 + 39 = 129
- Profit = 136 . 5 - 129 = \$ 7 . 5 â he starts earning money in year 13 .

Therefore , he starts earning money in year 13 .

< answer > 13 < /answer >